THE COPPERBELT UNIVERSITY
SCHOOL OF MATHEMATICS & NATURAL SCIENCES
CS235 & CS334 DATABASE TECHNOLOGY

02/10/2014

Instructions: Answer all the questions and be as precise as possible in your explanations

Time: 2hrs

Q 1 State and briefly discuss the four design guidelines for relational schemas. **[8 marks]**

Q 2 The table below shows the details of patient's appointments.

EmpNo	DocName	PatientNo	PatientName	Appointment		SurgeryNo
				Date	Time	
E1011	John Sakala	P100	Magi Mwale	03/04/14	10:00	S10
E1011	John Sakala	P105	Mat Banda	04/04/14	12:00	S15
E1024	Mary Chanda	P108	Tom Musonda	03/05/14	10:00	S10
E1024	Mary Chanda	P108	Tom Musonda	05/05/14	10:00	S10
E1032	Tom Moono	P105	Mat Banda	02/06/14	16:30	S15
E1032	Tom Moono	P110	Son Muntu	03/06/14	18:00	S13

a) The table above is susceptible to update anomalies. Provide examples from the table of insertion, deletion, and modification anomalies. **[6 marks]**

b) Describe and illustrate the process of normalizing the table above up to 3NF. **[6 marks]**

Q 3 Normalize the table below to BCNF. **[4 marks]**

Student	subject	Lecturer
1314000	Database	Dr Chaamwe
1314000	OOP	Mr Lengwe
1310234	Database	Dr Chaamwe
1310456	MIS	Mr Mumba
1311678	Internet Technology	Dr Libati

Q 4 State and describe the measures of correctness in fragmentation in distributed database systems. **[6 marks]**

Q 5 Consider the LIBRARY relational database schema shown in the given Figure and write down relational algebra expressions and SQL statements for the following queries. **[10 marks]**

i) List the details of borrowers who borrowed a book with the title "Introduction to databases"

(1) SQL } make algebra begin
normalisation - questions

1 fragmentation
2 ER diagram
3 from model relation schema

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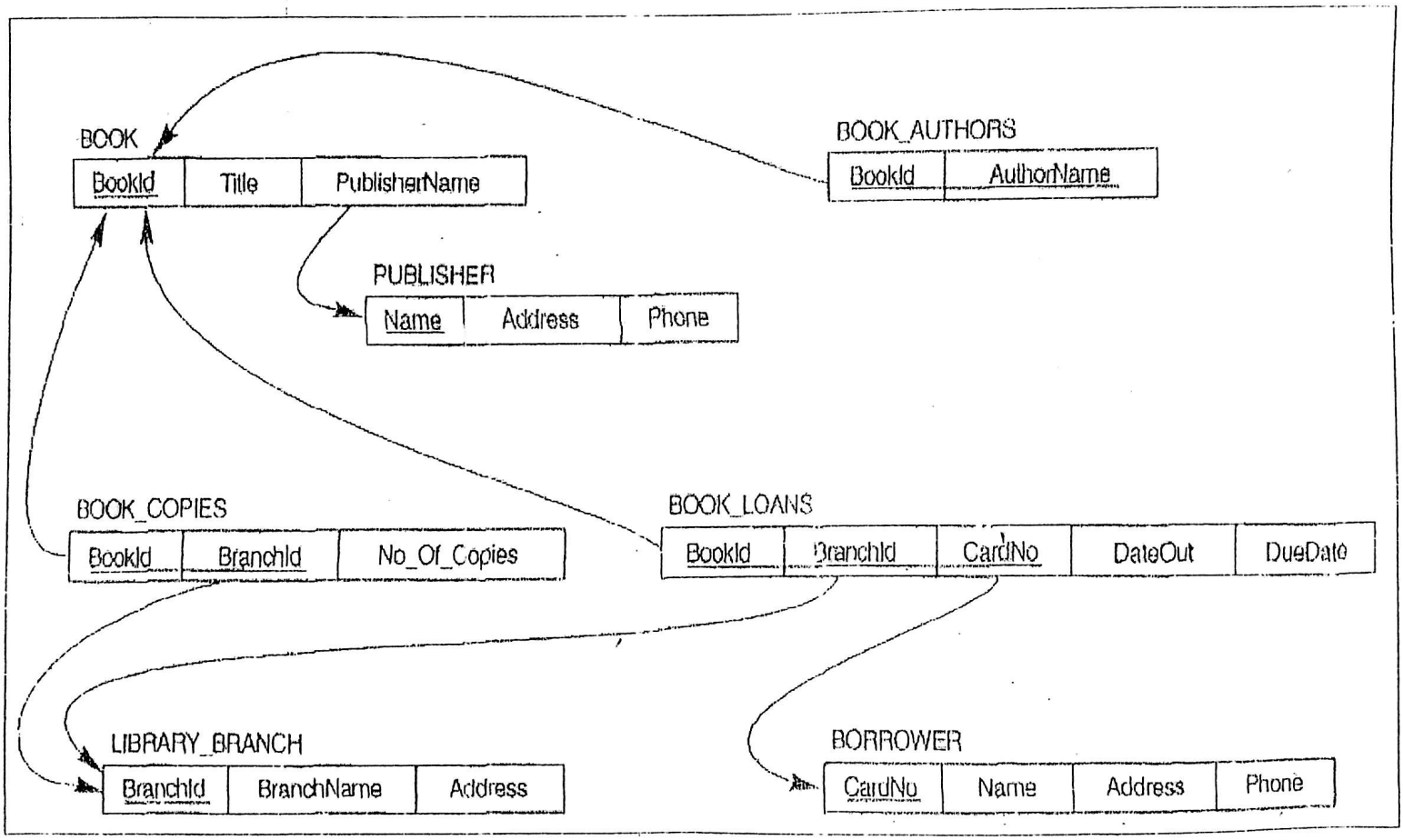
ii) List the details of a Branch library that have more than 100 copies of a book titled "Management Information Systems"

iii) For each book authored (or coauthored) by 'Almasri and Navethe,' retrieve the title and the number of copies owned by the library branch whose name is 'Kitwe City council Library.'

iv) For each book that is loaned out from the 'CBU Library' branch and whos DueDate is today, retrieve the book title, the borrower's name, and the borrower's address.

v) List the details of borrowers that borrowed books from CBU library on 3rd June 2014.

LIBRARY relational database schema



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